

AeroDesk - Top 25 Cheat Sheet

Last-Minute Revision - Most Likely Viva Questions

Project: Passenger Flight Booking Through Agency (AeroDesk)

Stack

1. What is your project? -> A web-based Airline Reservation & Booking Management System where agencies book flights for passengers; demonstrates DBMS concepts (schema, SQL, joins, normalization).
2. Tech stack? -> HTML, CSS, JavaScript (frontend), PHP (server), MySQL (database), XAMPP (local server).
3. How many tables? Name them. -> 6: passenger, agency, flight, booking, users, audit_logs.
4. Primary keys? -> passenger.pid, agency.aid, flight.fid, booking.bid, users.uid, audit_logs.lid.
5. Foreign keys? -> booking.pid -> passenger, booking.aid -> agency, booking.fid -> flight.
6. What is booking table? -> The relationship/junction table linking Passenger x Agency x Flight (resolves many-to-many).
7. Which normal form? -> 3NF (no partial or transitive dependencies).
8. Primary key vs Foreign key? -> PK uniquely identifies a row (not null/unique); FK references another table's PK for integrity.
9. What is normalization? -> Organizing data to remove redundancy and avoid insert/update/delete anomalies.
10. What is referential integrity? -> FK must match an existing PK - e.g. can't delete a flight that has bookings.
11. What is a JOIN? Example? -> Combines rows from tables. `SELECT p.pname FROM passenger p JOIN booking b ON p.pid=b.pid.`
12. WHERE vs HAVING? -> WHERE filters rows before grouping; HAVING filters groups after GROUP BY.
13. What is a subquery? -> A query inside another, e.g. `WHERE pid NOT IN (SELECT pid FROM booking).`
14. Aggregate functions? -> COUNT, SUM, AVG, MIN, MAX (used in reports & the 30 queries).
15. DELETE vs TRUNCATE vs DROP? -> DELETE removes rows (WHERE); TRUNCATE empties a table fast; DROP removes the table itself.
16. DDL vs DML? -> DDL = CREATE/ALTER/DROP (structure); DML = SELECT/INSERT/UPDATE/DELETE (data).
17. Architecture? -> Frontend (HTML/CSS/JS) -> PHP JSON API (api.php) via AJAX/fetch -> MySQL.
18. What are the user roles? -> Admin (full control), Agent (book/manage + reports), Customer (search/book + own ticket).
19. How is access controlled? -> Server-side role checks on every API action + role-filtered menu; not just hidden UI.
20. How are passwords stored? -> Hashed with PHP `password_hash()` (bcrypt), verified with `password_verify()`.
21. What are audit logs? -> Record of every sensitive action (login, book, cancel, CRUD, export) with user, role, IP, time.
22. What is the SQL Lab? -> A page that runs all 30 case-study queries live against MySQL and shows results.
23. Key features? -> Search & book, PDF boarding pass, CSV/Excel export, reports, CRUD, audit logs, responsive

glassmorphism UI.

24. How to run it? -> Put folder in xampp/htdocs, start Apache + MySQL, open localhost/aerodesk/home.php. Login: admin/admin123.
25. One-line summary? -> "A secure, role-based, responsive flight-booking web app demonstrating end-to-end DBMS concepts."

Demo logins: admin/admin123 - agent/agent123 - customer/customer123

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